

Activity

SDLC to Real Systems

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(S)TeemBang: Fitness Tracking Mobile App



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Part 2:

System Planning

A. Project Review

1. Project Title

(S)TeemBang: Fitness Tracking Mobile Application

2. Problem Statement

The growing demand for digital health monitoring solutions has led to widespread adoption of fitness applications. However, many existing platforms suffer from feature overload, poor user experience, limited personalization, and weak data analytics capabilities. Users often abandon fitness application due to lack of actionable insights, slow performance, or inconsistent tracking mechanism.

There is a need for scalable, secure, and performance-optimized mobile application that enables users to monitor physical activity, set measurable goals, visualize progress through analytics, and maintain long-term engagement. The proposed system aims to address these gaps through a modular, data-driven, and user-centric architecture.

3. Target Users

The target users for this mobile application are:

- Health-conscious individuals
- Student and working professionals
- Gym members and home-workout users
- Beginners starting fitness journey
- Individuals monitoring weight loss or physical conditioning

4. Project Scope

Included Features

- Secure user authentication (JWT/OAuth-based)
- Activity tracking (steps, workout sessions, calories burned)
- Goal setting (daily, weekly, monthly targets)
- Real-time dashboard analytics
- Historical data visualization (charts and trends)
- Profile management
- Local device sensor integration (basic pedometer API)
- Data persistence via cloud database

Excluded Features

- Wearable device API integration

- AI-based personal coaching
- Nutrition and meal planning engine
- Subscription/payment processing
- Telehealth or trainer booking services
- Social media sharing ecosystem

B. Requirement Identification

1. Functional Requirements

- FR1: The system shall provide secure user registration and authentication.
- FR2: The system shall allow users to log workouts and categorize activities.
- FR3: The system shall track daily step count via device sensors.
- FR4: The system shall calculate calories burned using fitness algorithms.
- FR5: The system shall allow users to define and monitor fitness goals.
- FR6: The system shall generate data analytics and visual reports.
- FR7: The system shall store historical activity logs securely.
- FR8: The system shall support push notifications and reminders.

2. Non-Functional Requirements

- NFR1: System response time must be less than 2 seconds.
- NFR2: System must support at least 10,000 concurrent users.
- NFR3: All data must be encrypted using TLS 1.2+.
- NFR4: System availability must be at least 99%.
- NFR5: Application must follow mobile-first responsive UI principles.
- NFR6: Codebase must follow clean architecture and modular design patterns.

Part 3:

SDLC Plan Phase

1. Planning

Activities:

- Business requirement analysis
- Stakeholder identification
- Technical feasibility assessment
- Risk assessment and mitigation planning
- Technology stack selection

Proposed Technology Stack:

- Frontend: Flutter / React Native
- Backend: Node.js (Express) or Laravel
- Database: PostgreSQL or MongoDB
- Cloud: AWS / Firebase
- Authentication: JWT-based system

Project Charter

Project Charter	
Title	Fitness Tracking Mobile App
Start Date	02/21/26
End Date	04/21/26
Project Manager	ZERRUDO, JASEKARL
System Analyst & Technical Writer	TUAZON, JOHN RAE
System Designer	APAS, CHRISTEL GRACE
System Developer	LABE, SHERWIN
Stakeholders	Project Team (Group 4) Future App Users (students, professionals, fitness beginners)
Project Objectives	The purpose of this project is to develop a mobile application called FitTrack that helps users track their daily fitness activities such as steps, workouts, calories burned, and goals. The app aims to help users stay motivated and improve their health by monitoring their progress.
Project Objectives	• Develop a working fitness tracking mobile app within 10 weeks • Allow users to register, login, and track activities • Provide progress reports and goal tracking • Ensure the system is secure and easy to use • Successfully deploy the app for testing and demonstration
Project Scope	
Project Timeline (Project Duration: 10 Weeks)	• Weeks 1-2: Planning • Week 3: Requirements • Weeks 4-5: Design • Weeks 6-8: Development • Week 9: Testing • Week 10: Deployment
Estimated Resources	
Risks	Delays in development Technical problems in coding Data security issues Lack of user feedback

Figure 1.1, (S)TeemBang: Fitness Tracking Mobile App Project Charter.

Gantt Chart

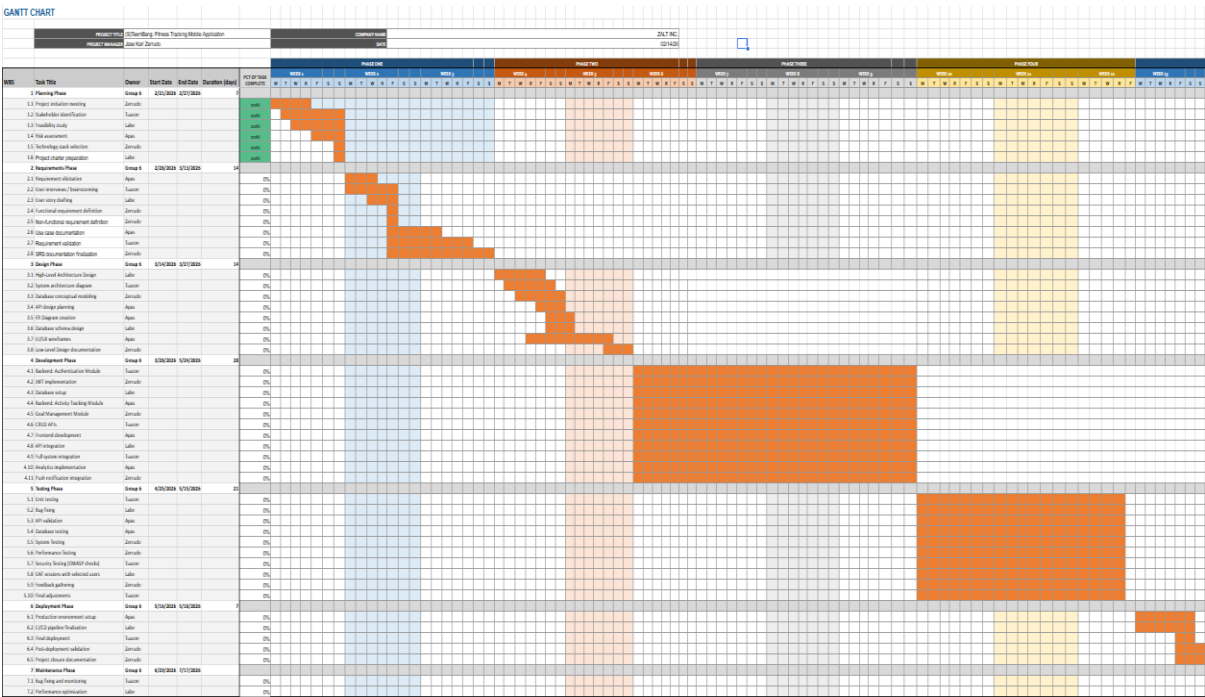


Figure 1.2, (S)TeemBang: Fitness Tracking Mobile App Gantt Chart.

Expected Deliverables

- Project Scope Document
- Risk Register
- High-Level Architecture Overview
- Project Timeline (Gantt Chart)

2. Requirements

Activities:

- Requirement elicitation workshops
- User story creation
- Functional decomposition
- Requirement validation

Deliverables:

- Software Requirements Specification (SRS)
- User Stories & Acceptance Criteria
- Use Case Diagrams

3. Design

Activities:

- System architecture design (Client-Server architecture)
- Database schema modeling (ERD)
- API endpoint design
- UI/UX wireframing
- Security architecture planning

Deliverables:

- High-Level Design (HLD)
- Low-Level Design (LLD)
- ER Diagram
- API Documentation
- UI Wireframes

4. Development**Activities:**

- Frontend module development
- Backend API development
- Database implementation
- Authentication & authorization implementation
- Integration of analytics modules

Deliverables:

- Functional application build
- Source code repository (Git-based)
- API implementation
- CI/CD pipeline configuration

5. Testing**Testing Types:**

- Unit Testing
- Integration Testing
- System Testing
- Performance Testing
- Security Testing (OWASP validation)
- User Acceptance Testing (UAT)

Deliverables:

- Test Plan
- Test Cases
- Bug Tracking Report
- UAT Sign-Off Document

6. Deployment**Deployment Strategy:**

- Cloud-based deployment (AWS EC2 / Firebase Hosting)
- CI/CD automated deployment pipeline
- Staging environment validation prior to production release
- Versioned release management

7. Maintenance**Future Enhancements:**

- AI-driven personalized workout recommendations
- Smartwatch API integration (Google Fit / Apple Health)
- Nutrition and hydration tracking
- Gamification system (badges and leaderboards)
- Machine learning-based performance prediction
- Multi-language support

SDLC Model Selection

Agile (Scrum Framework)

Why Agile is Most Suitable

The Fitness Tracking Mobile App requires iterative feature delivery, continuous improvement, and user feedback integration. Agile enables incremental releases, rapid prototyping, and adaptability to evolving requirements. Since mobile app usability and user engagement are critical success factors, frequent sprint reviews and refinements are essential.

Agile supports:

- Continuous integration and deployment
- Early validation of UI/UX
- Risk reduction through short development cycles
- Stakeholder involvement at every sprint

Why Other Models Are Less Suitable

Model	Reason
Waterfall	Rigid structure; limited flexibility for UI/UX improvements once development begins.
Spiral	Risk-driven and suitable for large-scale, high-risk enterprise systems; excessive for this scope.
RAD	Requires extensive prototyping resources and highly experienced developers.
DevOps (standalone model)	More of a culture/practice than a pure SDLC model; complements Agile but does not replace structured SDLC planning.
Big Bang	High risk due to lack of formal planning and structured validation.
Iterative	Effective, but Agile provides better sprint structure, backlog prioritization, and stakeholder collaboration.

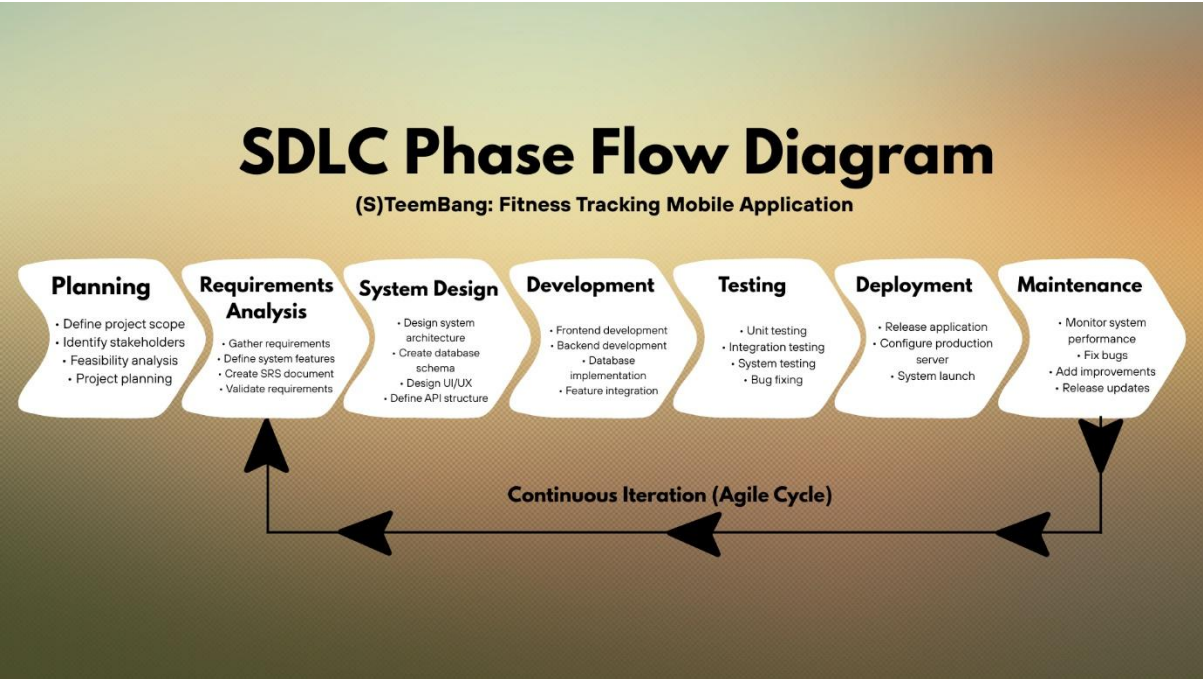
Risks Reduced by Agile

- Requirement volatility risk
- Late-stage defect discovery
- Stakeholder misalignment
- Feature irrelevance
- Budget overruns due to rework

Visual Diagram

(S)TeemBang: Fitness Tracking Mobile App

SDLC Phase Flow Diagram



Model Representation

